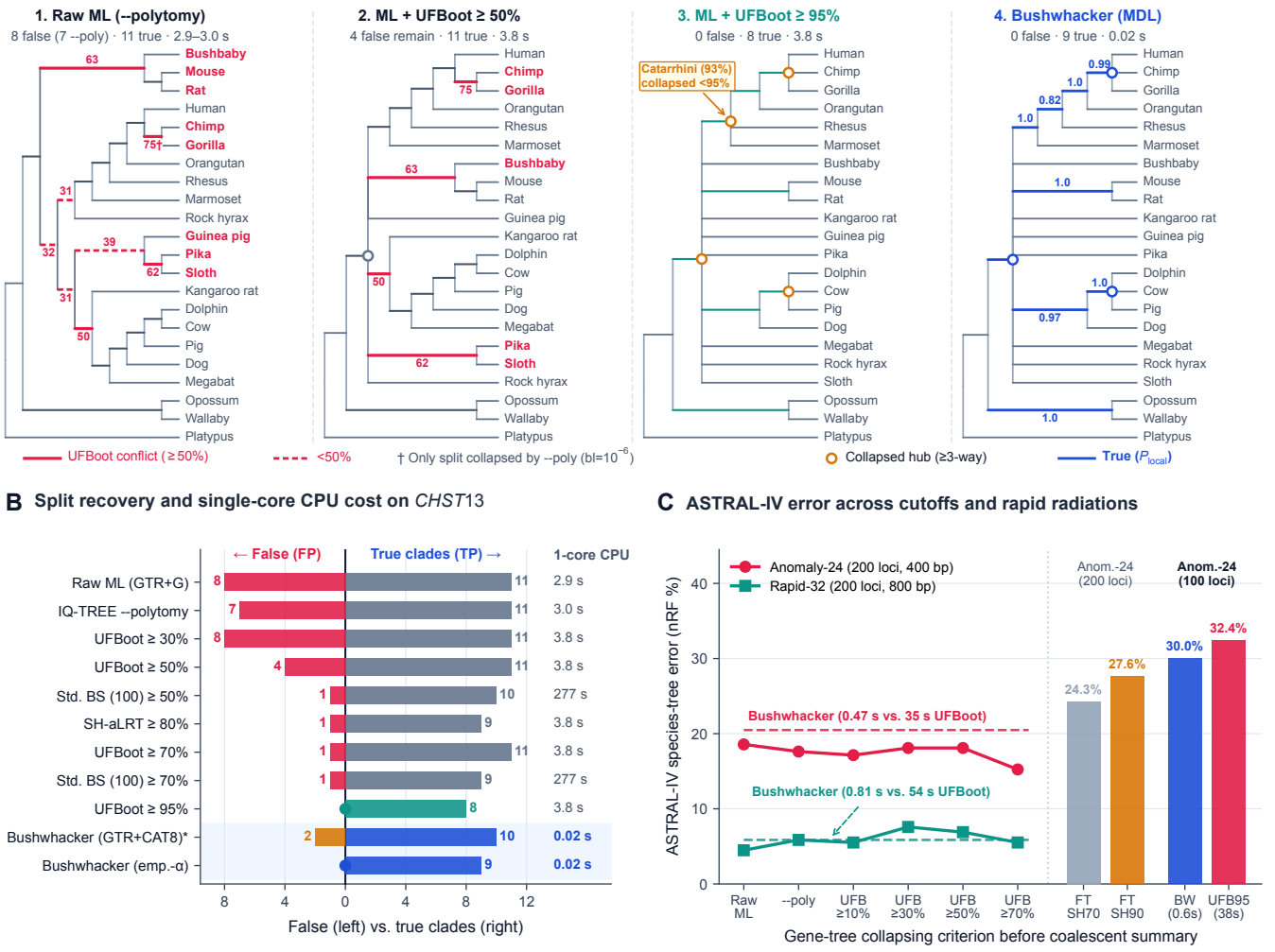

A CHST13 topologies across support thresholds and Bushwhacker MDL (22 mammals, 1,176 bp)
1. Raw ML (--polytomy)
8 false (7 --poly) · 11 true · 2.9–3.0 s
2. ML + UFBoot ≥ 50%
4 false remain · 11 true · 3.8 s
3. ML + UFBoot ≥ 95%
0 false · 8 true · 3.8 s
4. Bushwhacker (MDL)
0 false · 9 true · 0.02 s
Catarrhini (93%) collapsed <95%
UFBoot conflict (≥ 50%)
<50%
† Only split collapsed by --poly (bl=10−6)
Collapsed hub (≥3-way)
True (Plocal)
B Split recovery and single-core CPU cost on CHST13
← False (FP)
True clades (TP) →
1-core CPU
Raw ML (GTR+G)
IQ-TREE --polytomy
UFBoot ≥ 30%
UFBoot ≥ 50%
Std. BS (100) ≥ 50%
SH-aLRT ≥ 80%
UFBoot ≥ 70%
Std. BS (100) ≥ 70%
UFBoot ≥ 95%
Bushwhacker (GTR+CAT8)*
Bushwhacker (emp.-α)
False (left) vs. true clades (right)
C ASTRAL-IV error across cutoffs and rapid radiations
Anomaly-24 (200 loci, 400 bp)
Rapid-32 (200 loci, 800 bp)
Anom.-24 (200 loci)
Anom.-24 (100 loci)
Bushwhacker (0.47 s vs. 35 s UFBoot)
Bushwhacker (0.81 s vs. 54 s UFBoot)
ASTRAL-IV species-tree error (nRF %)
Raw ML
--poly
UFB ≥10%
UFB ≥30%
UFB ≥50%
UFB ≥70%
FT SH70
FT SH90
BW (0.6s)
UFB95 (38s)
Gene-tree collapsing criterion before coalescent summary